

Multi-machine & Parallel manufacturing

MOs, work orders, production lines, and machines work together — including split operations for parallel extraction and processing.

PVS ERP · Manufacturing module

This document explains how production lines and machines are assigned to manufacturing orders (MOs) and work orders (WOs), and how the same MO can run **in parallel on multiple machines** without picking “multiple machines” on a single work order.

Related docs: [manufacturing-multi-step-bom.md](#) · [soap-manufacturing-process.md](#) · [erp-user-manual.md](#)

1. Design decision (read this first)

Question	Answer in PVS ERP
Can one work order run on several machines at once?	No. Each WO has exactly one <code>lineId</code> and one <code>machineId</code> .
Can one MO run on several machines at once?	Yes , by creating several work orders for the same routing step, each on a different line/machine with its own quantity.
How do I split qty across machines?	Use Split operation on the MO (UI) or <code>POST .../split-operation</code> (API).
Should we add “select multiple machines” on the MO?	Not recommended. Tracking status, output, QA, and machine idle/busy state requires one machine per WO .

This follows Odoo MRP: parallel capacity is modeled as **multiple work orders per operation**, not one WO bound to many machines.

2. Concept map

```
flowchart TB
    subgraph facility [Production facility e.g. WC-OIL]
        L1[Line WC-OIL-EXT-01]
        L2[Line WC-OIL-EXT-03]
        L3[Line WC-OIL-FLT-01]
    end
    end
    L1 --> M1[Machine MCH-OIL-EXT-01]
    L2 --> M2[Machine MCH-OIL-EXT-03]
    MO[Manufacturing order MO-2026-0042<br/>plannedQty 600 L]
    MO --> W01[WO MO-.../1a · Extract · 360 L<br/>EXT-01 · MCH-01]
    MO --> W02[WO MO-.../1b · Extract · 240 L<br/>EXT-03 · MCH-03]
    MO --> W03[WO MO-.../2 · Filter · 600 L<br/>waiting until 1a+1b done]
    W01 --> M1
    W02 --> M2
    W03 --> L3
```

Entity	Role	Parallel relevance
Production facility	Plant / work center group (e.g. Oil Extraction WC-OIL)	MO is created for one facility
Production line	Logical work center (e.g. WC-OIL-EXT-01)	Each parallel WO picks one line
Machine	Physical asset on a line (e.g. MCH-OIL-EXT-01)	Each parallel WO picks one machine
BOM operation	Routing step on the BOM (Extract, Filter, Fill)	Split applies per operation
Manufacturing order (MO)	Batch header: product, planned qty, dates	One MO number; qty split across WOs
Work order (WO)	Executable step on the shop floor	Unit of parallel execution

3. Data model

3.1 Key tables

ProductionOrder (MO)

Field	Meaning
plannedQty	Total batch quantity
facilityId	Where this MO runs
lineId	Optional MO-level default line (legacy/summary); per-WO line is authoritative for execution

WorkOrder (WO)

Field	Meaning
workOrderNo	e.g. MO-2026-0042/1a — suffix a , b , ... when split
bomOperationId	Which routing step this WO belongs to
target	Qty this WO must produce
plannedSplitQty	Same as target when created via split
splitSeq	0 = unsplit default; 1, 2, ... = parallel siblings
lineId	Assigned production line
machineId	Assigned machine (optional until start)
status	waiting → ready → running → complete (+ QA)

BomOperation

Field	Meaning
seq	Step order (1, 2, 3...)
blockedByOperationId	Predecessor operation (Filter blocked by Extract)
lineId / machineId	Default line/machine when MO spawns WOs
requiresQa	If true, WO stays QA-pending until supervisor approves

BomOperationLine (eligible lines)

Lists which production lines **may** run an operation (e.g. Extract allowed on EXT-01 ... EXT-06). Used in BOM editor; split UI shows all active lines in the facility.

Machine

Belongs to exactly one productionLineId . A WO's machine must belong to its chosen line.

3.2 Work order numbering after split

For operation seq 1 with three parallel splits:

workOrderNo	splitSeq	Qty
MO-xxx/1a	0	200
MO-xxx/1b	1	200
MO-xxx/1c	2	200

Suffix letters are assigned in split order (a ... z).

4. Two ways to run in parallel

4.1 Option A — Split operation (one MO, multiple WOs)

Best when: One batch number, one material issue, downstream step waits for **all** parallel extract WOs.

Flow:

1. Create MO (e.g. 600 L groundnut extract).
2. Release MO and issue materials.
3. Open MO → **Work orders** tab.
4. Click **Split Extract** (or the operation name).
5. In the modal, add rows:

Line	Machine	Qty
WC-OIL-EXT-01	MCH-OIL-EXT-01	360
WC-OIL-EXT-03	MCH-OIL-EXT-03	240

6. **Apply split** — backend replaces the single Extract WO with two WOs (/1a , /1b).
7. **Start** each WO independently when ready.
8. **Filter** WO remains `waiting` until **both** extract WOs are `complete` and QA passed (if required).

Rule: Split quantities **must sum exactly** to `MO.plannedQty` .

4.2 Option B — Multiple MOs (seed-press oil today)

Best when: Each stage is already a **separate BOM/MO** (Extract MO → Filter MO → Pack MO).

Example for 600 L unfiltered target:

MO	BOM	Line	Qty
MO-A	Rev-Oil-Extract-1.0	WC-OIL-EXT-01	360 L
MO-B	Rev-Oil-Extract-1.0	WC-OIL-EXT-03	240 L

Filter MO consumes **pooled** semi-finished stock from both extract MOs. No split modal needed — parallelism is at **MO level**.

Both options are valid; choose based on whether you want one MO number or separate extract batches.

5. Single machine assignment (no split)

When only **one** line/machine runs a step, use inline dropdowns on the Work orders tab (or **Assign line / machine** in the MO header).

Per work order:

1. **Production line** — required before **Start**
2. **Machine** — optional but recommended when multiple machines exist on the line

Saves via:

```
PATCH /v1/production-orders/{moId}/work-orders/{woId}/assign
Content-Type: application/json

{
  "lineId": "<production-line-id>",
  "machineId": "<machine-id-or-null>"
}
```

Bulk assign (same line for all open WOs, with optional per-WO overrides):

```
PATCH /v1/production-orders/{moId}/assign-line
Content-Type: application/json

{
  "lineId": "<production-line-id>",
  "workOrderAssignments": [
    { "workOrderId": "...", "lineId": "...", "machineId": "..." }
  ]
}
```

Constraints:

- Cannot reassign line/machine while WO is `running` or `complete`.
- Machine must belong to the selected line.
- Line must belong to the MO's facility.

6. Split operation — backend behaviour

Endpoint:

```
POST /v1/production-orders/{moId}/split-operation
Content-Type: application/json

{
  "bomOperationId": "<operation-id>",
  "splits": [
    { "lineId": "...", "machineId": "...", "qty": 360 },
    { "lineId": "...", "machineId": "...", "qty": 240 }
  ]
}
```

Algorithm (`backend/src/lib/mo-work-orders.ts` → `splitOperationWorkOrders`):

1. Validate operation exists.
2. Validate `sum(splits.qty) === MO.plannedQty`.
3. **Delete** all existing WOs for that `bomOperationId` on this MO.
4. **Create** one WO per split row with `lineId`, `machineId`, `target`, `plannedSplitQty`, `splitSeq`.
5. Set initial status to `waiting` if the operation has a blocked-by predecessor, else `ready`.
6. Call `refreshWaitingReady` to unblock downstream steps when predecessors finish.

Important: Splitting **replaces** WOs for that operation only. Other operations (Filter, Fill) are untouched. Do not split after WOs are already running or complete.

7. Dependencies & QA with parallel WOs

7.1 Operation dependencies

If operation **Filter** is blocked by **Extract**:

- All WOs with `bomOperationId = Extract` must reach `complete` (+ QA pass if `requiresQa`).
- Only then does the Filter WO move from `waiting` → `ready`.

With splits `1a` and `1b`:

```
Extract 1a (running) ┘
Extract 1b (ready)  └─┬─> both complete + QA ─> Filter (ready)
```

7.2 QA failure

If QA fails on a WO:

1. Failed WO → `rework`, `qaStatus = fail`.
2. **Predecessor operation** WOs reopened to `ready` (all parallel extract WOs if filter failed downstream).
3. Later operations set back to `waiting`.

Parallel splits share the same dependency rules as a single WO — **all** predecessor WOs for that operation must be done.

8. Work order lifecycle

```
stateDiagram-v2
    [*] --> waiting: predecessor incomplete
    waiting --> ready: all predecessor WOs done + QA pass
    ready --> running: operator Start
    running --> complete: operator Done
    complete --> pending_qa: requiresQa
    pending_qa --> pass: QA approve
    pending_qa --> rework: QA fail
    rework --> ready: predecessor reopened
    pass --> [*]
```

Start requirements:

- Status must be `ready` (or legacy `queued`).
- `lineId` must be set (UI disables Start until line is chosen).

Each parallel WO is started and completed **independently**.

9. UI reference

9.1 Manufacturing → MO detail → Work orders

Control	Purpose
Production line / Machine dropdowns	Assign one line + machine to this WO
Start	Begin this WO on its machine
Done	Mark step complete
QA	Approve/reject when operation requires QA
Split {operation}	Open split modal for parallel lines

9.2 Split operation modal

- One row per parallel line.
- Fields: **Line, Qty, Machine (optional)**.
- **Add line** for 3+ parallel runs (e.g. six extractors).
- **Total** must match MO planned qty.

9.3 Assign line / machine (header button)

Shown when any open WO lacks a line. Opens modal to assign line + machine for multiple WOs at once.

10. Oil extraction example (WC-OIL)

Configured by `npm run db:configure-oil-extraction:dev` :

Line code	Role	Typical machine
WC-OIL-EXT-01 ... EXT-06	Parallel extraction	MCH-OIL-EXT-01 ... 06
WC-OIL-FLT-01 ... FLT-03	Filtering	MCH-OIL-FLT-01 ... 03
WC-OIL-FILL	Bottling / pouches	MCH-OIL-FILL

Scenario: MO for 600 L extract, run on two presses in parallel.

Using split (multi-op BOM on one MO):

1. MO planned qty = 600.
2. Split Extract → 360 on EXT-01, 240 on EXT-03.
3. Start both WOs same day.
4. When both done (+ QA), Filter WO becomes ready for full 600 L.

Using separate MOs (current seed-press BOM layout):

1. MO-1: 360 L on EXT-01.
 2. MO-2: 240 L on EXT-03.
 3. Create Filter MO when enough unfiltered stock is available.
-

11. API summary

Method	Path	Purpose
POST	/production-orders	Create MO; spawns one WO per BOM operation
PATCH	/production-orders/:id/assign-line	MO-level + bulk WO line/machine assign
PATCH	/production-orders/:id/work-orders/:woId/assign	Single WO line/machine assign
POST	/production-orders/:id/split-operation	Parallel qty split for one operation
POST	/production-orders/:id/work-orders/:woId/start	Start WO
POST	/production-orders/:id/work-orders/:woId/done	Complete WO
POST	/production-orders/:id/work-orders/:woId/qa	QA pass/fail
GET	/production-lines?facilityId=	Lines for dropdowns
GET	/machines?facilityId=	Machines for dropdowns

12. Code locations

Area	Path
WO create / split / assign / QA	<code>backend/src/lib/mo-work-orders.ts</code>
HTTP routes	<code>backend/src/routes/manufacturing.ts</code>
Prisma models	<code>backend/prisma/schema.prisma</code> (<code>WorkOrder</code> , <code>BomOperation</code> , <code>BomOperationLine</code> , <code>Machine</code>)
Work orders UI	<code>erp-portal/src/components/manufacturing/MoWorkOrdersPanel.tsx</code>
Split modal	<code>erp-portal/src/components/manufacturing/SplitOperationModal.tsx</code>
Line/machine fields	<code>erp-portal/src/components/manufacturing/WoLineMachineFields.tsx</code>
Bulk assign modal	<code>erp-portal/src/components/manufacturing/AssignLineModal.tsx</code>
Oil lines + machines seed	<code>backend/ops-scripts/06-configure-oil-extraction.ts</code>
Site layout constants	<code>backend/ops-scripts/config/site-layout.ts</code>

13. Limitations & future work

Topic	Current behaviour
Multi-machine on one WO	Not supported by design
Split after start	Not allowed (WOs deleted/recreated on split)
Per-step material issue	All BOM components issued at MO release; per-operation issue is planned (see multi-step BOM doc)
Machine busy/idle	Updated on WO start/complete when <code>machineId</code> is set
Legacy BOMs (no operations)	Single WO <code>/1</code> ; split requires at least one <code>BomOperation</code>
Eligible lines enforcement	BOM stores eligible lines; split UI currently lists all active facility lines

14. Quick decision guide

Need parallel machines on the SAME batch (one MO number)?
↳ Use Split operation → multiple WOs, one machine each

Need parallel extract but separate batch tracking?
↳ Create multiple extract MOs on different lines

Need one machine on one step only?
↳ Use Work orders dropdowns (no split)

Wondering about "pick 3 machines on the MO"?
↳ Don't – use Split operation with 3 rows instead

15. Glossary

Term	Meaning
MO	Manufacturing order — batch header
WO	Work order — shop-floor executable unit
Split	Replace one operation's WO(s) with several parallel WOs
Operation	BOM routing step (Extract, Filter, ...)
Line	Production line / work center
Blocked by	Operation or WO that must finish before the next starts

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